

Kian Maleki — Data Scientist | Machine Learning | MLOps | Physics PhD

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University of Iowa (2026) PhD in Physics (GPA: 4.03) | Creighton University (2017) MS in Physics (GPA: 4.00)

TECHNICAL EXPERIENCE

TripleTen | AI & Machine Learning Program 2026

- Supervised & unsupervised learning (Scikit-learn) for real-world tasks.
- Deep learning: computer vision (image classification), NLP, and transformer-based workflows.
- End-to-end ML projects (15+) + tools: Git, SQL.

Extern, Pfizer | AI-Powered Document Intelligence Externship 2026

- Built an AI-powered document intelligence pipeline to automate processing of healthcare documents using native PDF text extraction, advanced OCR workflows, LLMs, and RAG architectures.
- Developed retrieval-augmented pipelines to improve accuracy and reduce hallucinations in document-based question answering systems.
- Delivered a functional prototype demonstrating how AI-driven automation improves efficiency in healthcare supply chain operations.

TECHNICAL EXPERTISE

- **Modeling:** Gen AI (RAG), Experiment Tracking (MLflow), Data Versioning (DVC), Drift Monitoring (Evidently, PSI), HPC (MPI/Multithreading), Monte Carlo Simulations, Bayesian Statistics.
- **Engineering:** Python (Expert), SQL, FastAPI, Pydantic, Docker, Linux/Bash, Version Control (Git), CI/CD (GitHub Actions, Jenkins).
- **Numeric Stack:** Pandas, NumPy, Polars, Numba, Scikit-Learn, Transformers, TensorFlow, PyTorch.
- **DevOps:** Automated Testing (pytest), Pydantic validation, AWS (S3, EC2), System Health Monitoring, Jenkins Pipelines.

DATA-DRIVEN PROJECTS

RAG-Powered Knowledge Assistant 2026

Tech: Python, PyTorch, Transformers, NLP, Docker

link: <https://github.com/drkianmaleki/RAG-Powered-Knowledge-Assistant>

- Built a local Retrieval-Augmented Generation (RAG) system for document question answering, enabling context-aware responses using Transformers and PyTorch.
- Improved answer relevance by designing a sliding-window chunking strategy with overlap, preserving semantic continuity across large documents.
- Containerized the application with Docker, enabling reproducible deployment and scalable local execution for NLP workloads.

Heart Disease Prediction: End-to-End MLOps Pipeline 2026

Tech: DVC, Git, MLflow, PyTest, GitHub Action, Evidently

link: <https://github.com/drkianmaleki/Heart-Disease-Prediction>

- Built an end-to-end MLOps pipeline for heart disease prediction, enabling reproducible data versioning, experiment tracking, and model registry using DVC and MLflow.
- Automated CI/CD workflows with GitHub Actions and pytest, enforcing performance thresholds (Accuracy > 0.80) prior to deployment.
- Improved model reliability by implementing data and concept drift monitoring with Evidently, enabling proactive retraining triggers.

PDF Document Intelligence Extractor (Pfizer Externship Project) 2026

Tech: Python, NLP, Regex, Pandas, PDF Text Extraction, RAG

link: <https://github.com/drkianmaleki/PDF-Document-Intelligence-Extractor>

- Built a document intelligence pipeline to extract key regulatory information from PDF documents using native text-layer parsing rather than OCR.
- Developed rule-based and NLP extraction methods to identify, normalize, and structure information from semi-structured formatted text.

- Converted unstructured document content into structured datasets, enabling downstream analysis and reducing manual review effort.

Ambiguity Range Framework

2026

Tech: Python, Scikit-Learn, XGBoost, PyTest, Statistical Testing, Model Evaluation

link: <https://github.com/drkianmaleki/ambiguity-framework>

- Developed a model-agnostic diagnostic toolkit for evaluating binary classifiers beyond global metrics such as AUC-ROC.
- Implemented ambiguity metrics to quantify prediction concentration near decision boundaries and threshold-sensitive degradation in precision, recall, and F1.
- Built reproducible experiments across real-world datasets with calibration analysis, bootstrap confidence intervals, paired statistical tests, and automated test coverage.

PROFESSIONAL EXPERIENCE

Creighton University | Assistant Professor of Physics

Aug 2025 – Present

- Teach undergraduate physics courses (40–180 students), delivering lectures on optics and electronics with a focus on applied problem-solving.
- Designed and implemented course curricula and syllabi, aligning learning objectives with departmental standards.
- Supervised and mentored 1–5 teaching assistants, improving instructional quality and ensuring consistency in grading.
- Developed interdisciplinary lab curriculum integrating AI tools and chatbots, teaching students how to use emerging technologies ethically in scientific workflows.

University of Iowa | Graduate Research Assistant (Physics)

Aug 2018 – Aug 2025

- Optimized large-scale numerical simulations, reducing computational runtime from days to under one hour through algorithmic improvements.
- Developed high-dimensional modeling frameworks for complex physical systems, applying advanced statistical and numerical methods.
- Analyzed high-dimensional experimental data to extract signal from noise, applying statistical inference and optimization techniques.

Creighton University | Graduate Research Assistant (Physics)

Aug 2015 – Aug 2018

- Developed simulation-based models for physical systems using Python and Fortran, including N-body simulations and molecular dynamics.
- Engineered parallelized computation pipelines for large-scale simulations, leveraging HPC resources to improve scalability.
- Built interactive scientific software tools (Python + GUI) for simulation and analysis, enabling user-driven exploration.

SELECTED PUBLICATIONS

- *Maleki, K. et al.* “Crystal fields, exchange and dipolar interactions...” **Phys. Rev. B**, 2025.
- *Maleki, K. et al.* “A General and Modular Approach to Solid-State Integration...” **Nano Letters**, 2025. (Related to a patented quantum device)